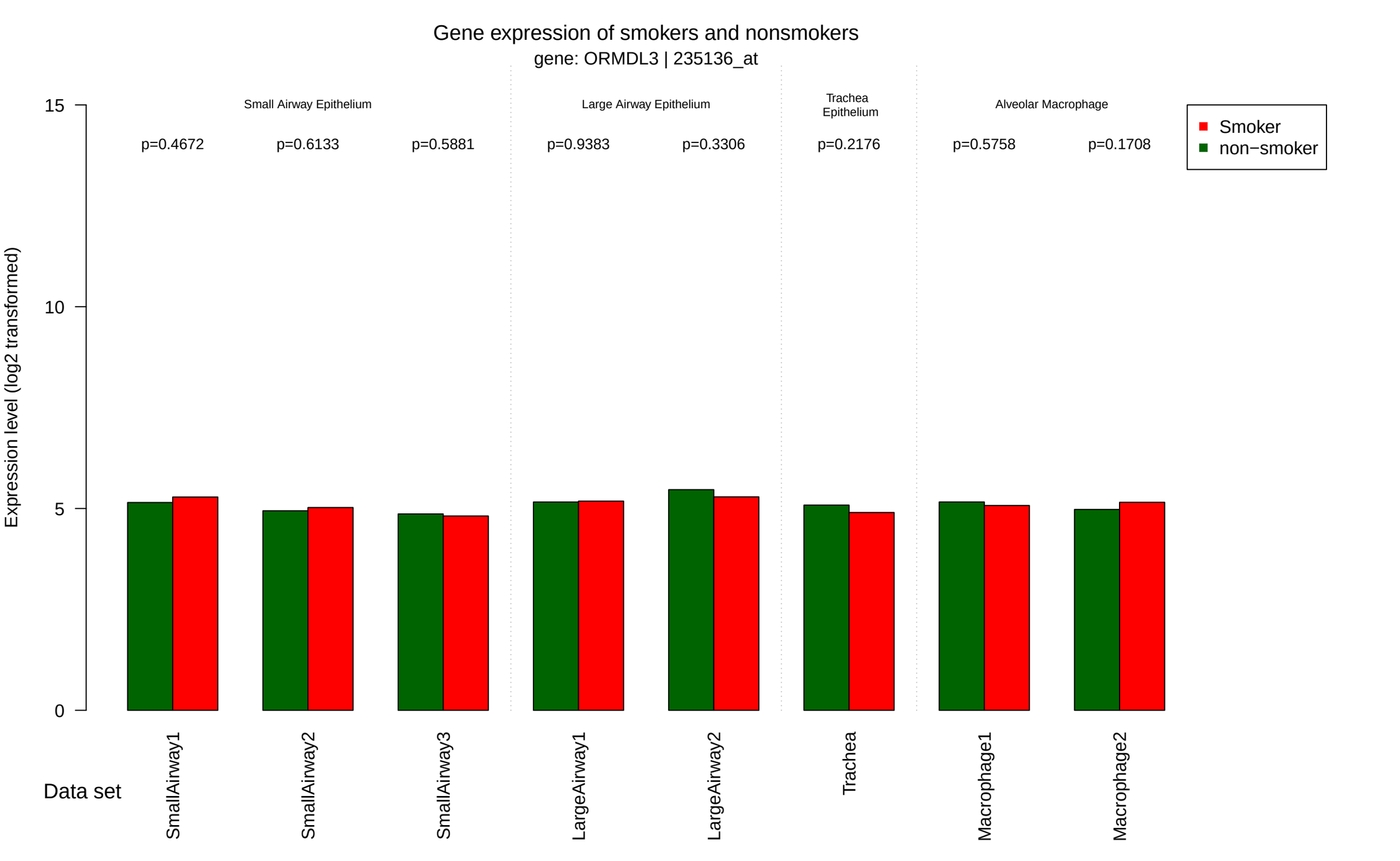
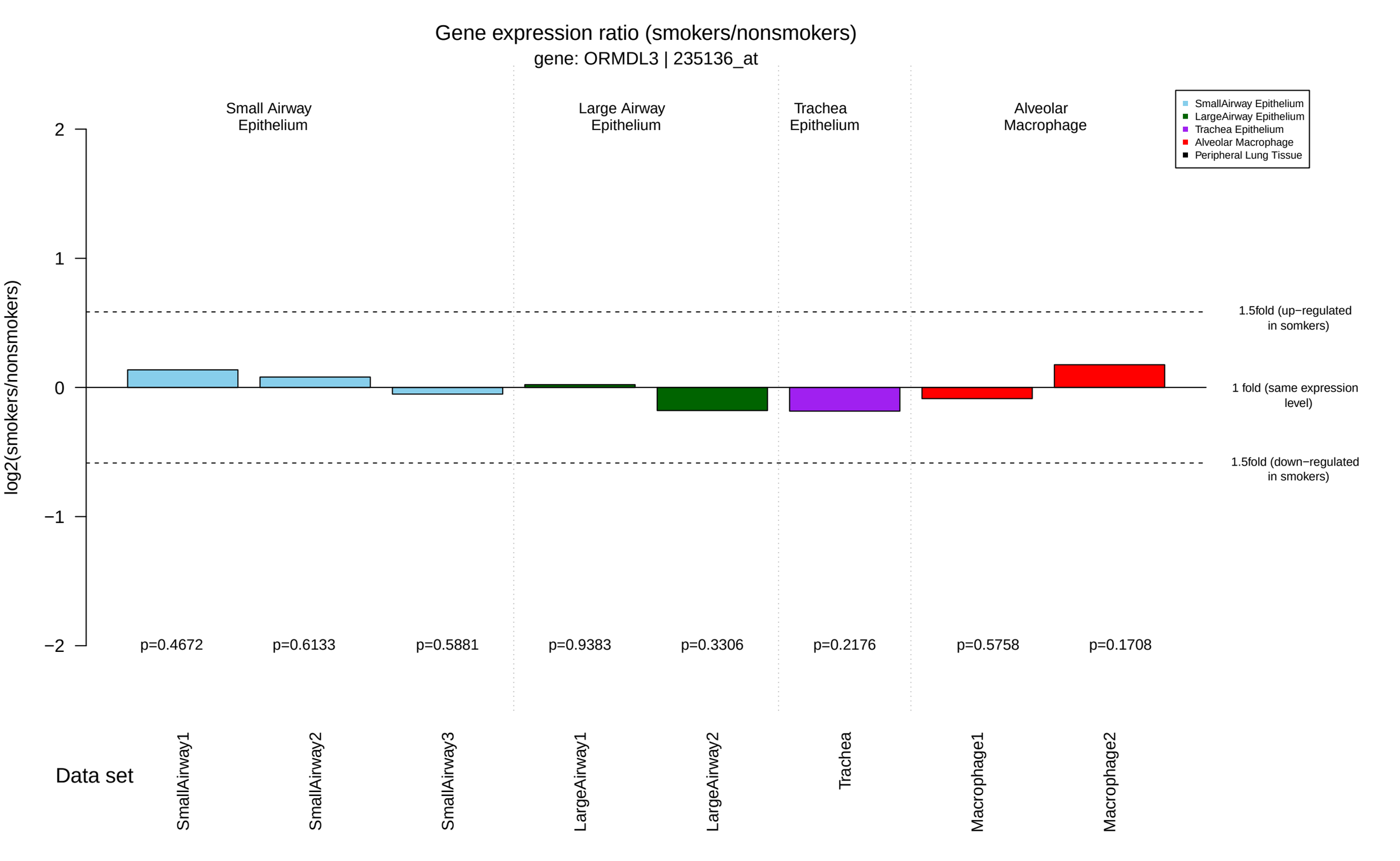
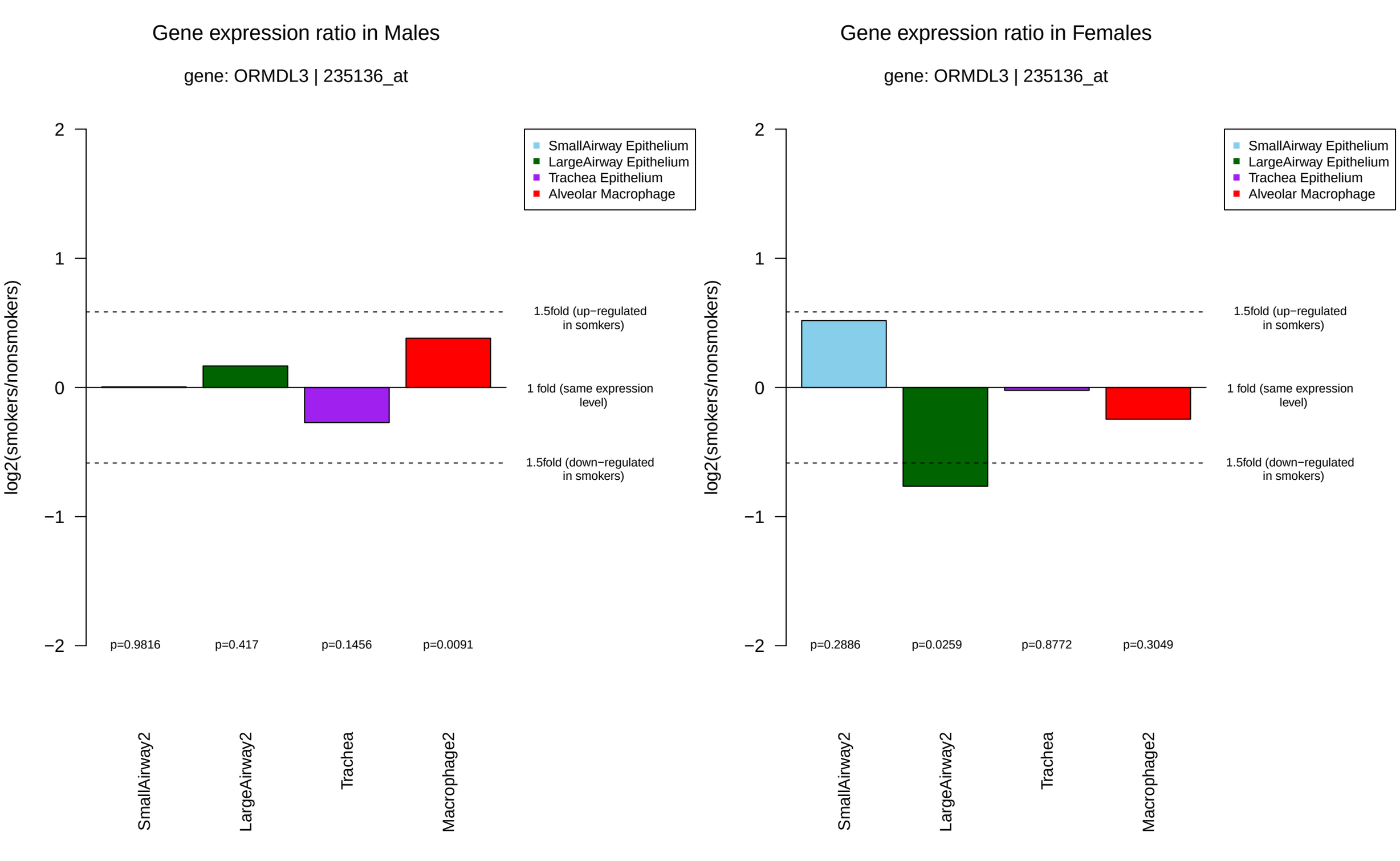
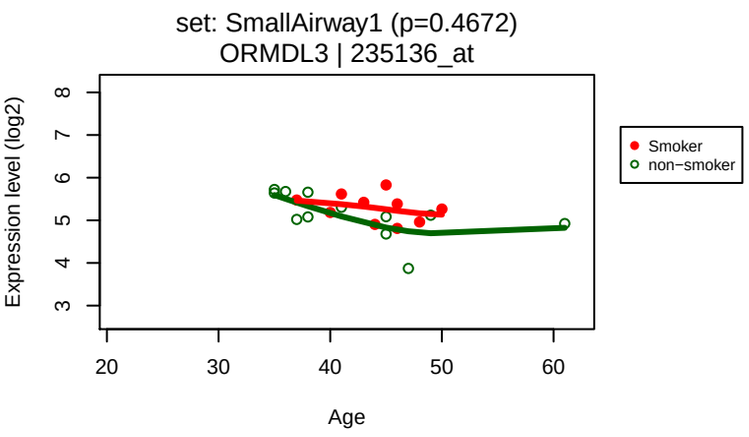
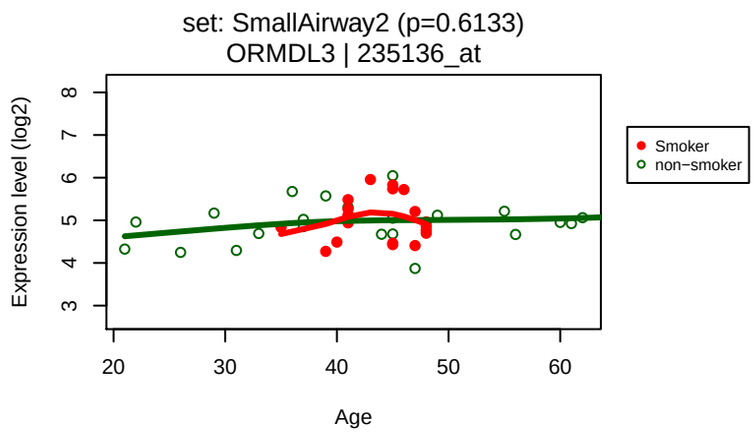
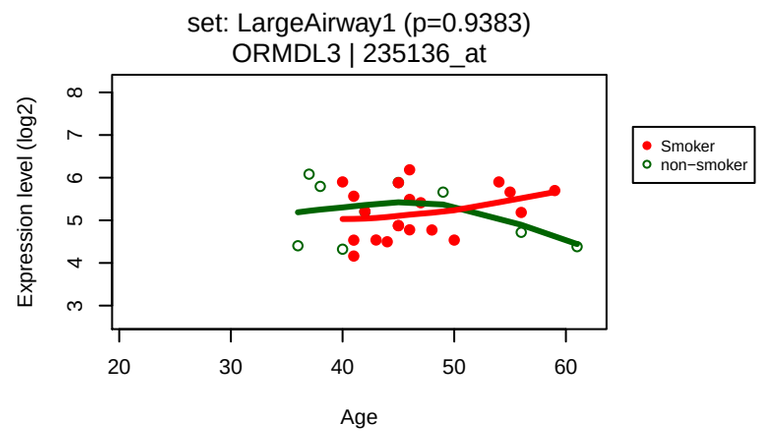

Gene expression of smokers and nonsmokers
gene: ORMDL3 | 235136_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4672
p=0.6133
p=0.5881
p=0.9383
p=0.3306
p=0.2176
p=0.5758
p=0.1708
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

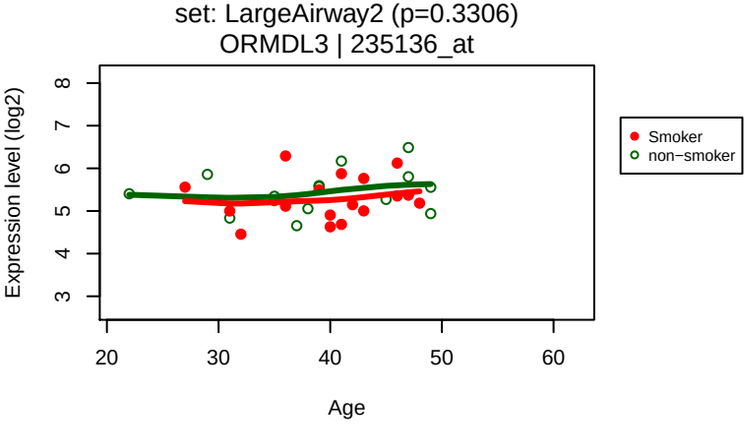

Gene expression ratio (smokers/nonsmokers)
gene: ORMDL3 | 235136_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4672
p=0.6133
p=0.5881
p=0.9383
p=0.3306
p=0.2176
p=0.5758
p=0.1708
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

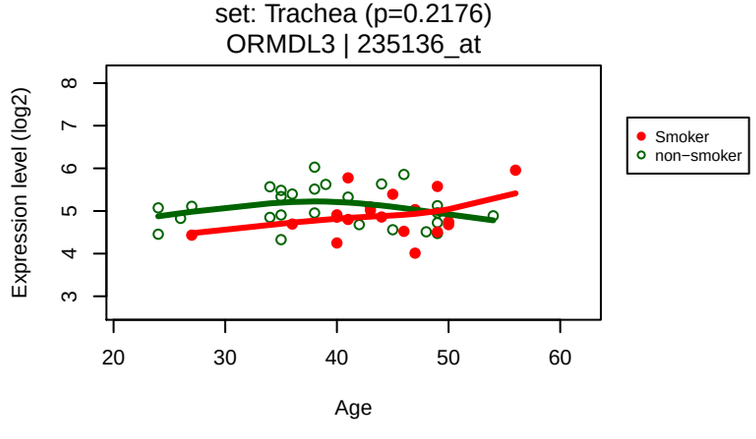

Gene expression ratio in Males
gene: ORMDL3 | 235136_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9816
p=0.417
p=0.1456
p=0.0091
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ORMDL3 | 235136_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2886
p=0.0259
p=0.8772
p=0.3049
SmallAirway2
LargeAirway2
Trachea
Macrophage2

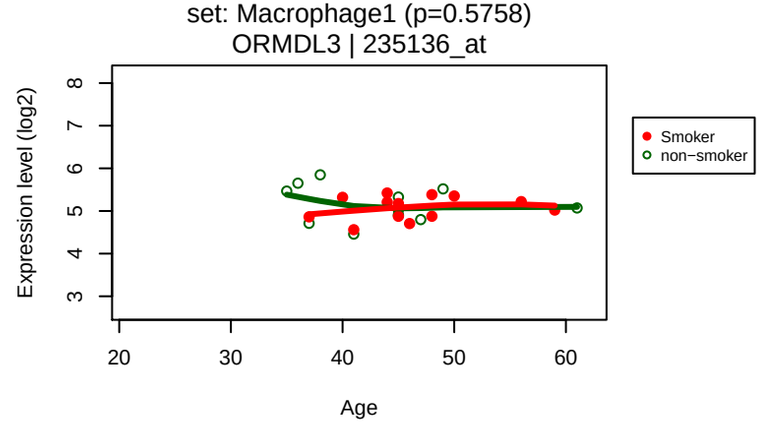

set: SmallAirway1 (p=0.4672)
ORMDL3 | 235136_at
Expression level (log2)
Age
Smoker
non−smoker

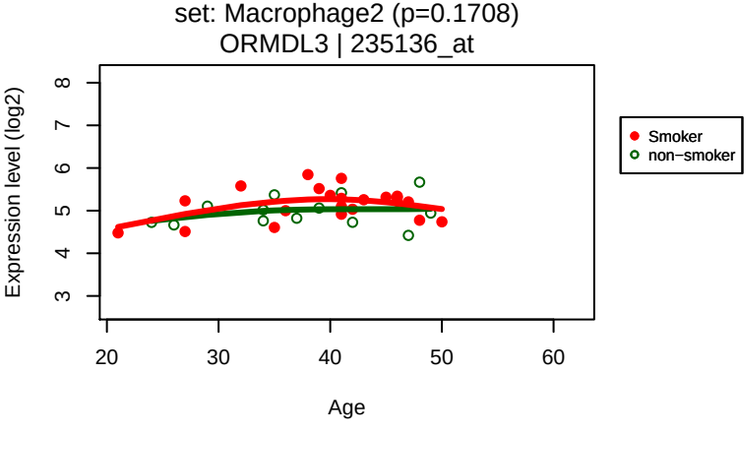

set: SmallAirway2 (p=0.6133)
ORMDL3 | 235136_at
Expression level (log2)
Age
Smoker
non−smoker

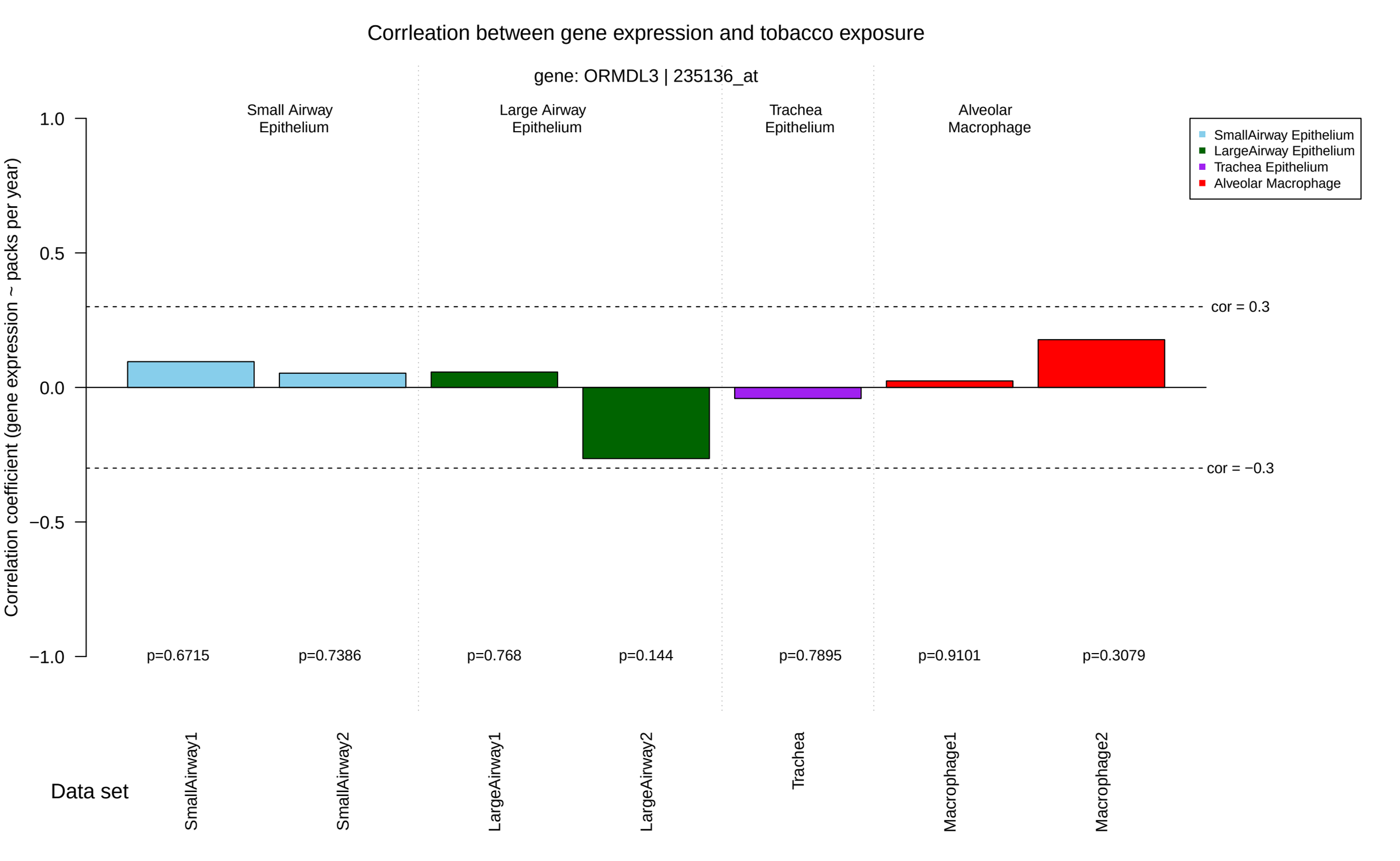

set: LargeAirway1 (p=0.9383)
ORMDL3 | 235136_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3306)
ORMDL3 | 235136_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2176)
ORMDL3 | 235136_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5758)
ORMDL3 | 235136_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1708)
ORMDL3 | 235136_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ORMDL3 | 235136_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6715
p=0.7386
p=0.768
p=0.144
p=0.7895
p=0.9101
p=0.3079
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2